

BABU BANARASI DAS
UNIVERSITY

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SCHOOL OF COMPUTER APPLICATIONS

NO SQL PROJECT ON
Student Management System Using MongoDB

Submitted to:

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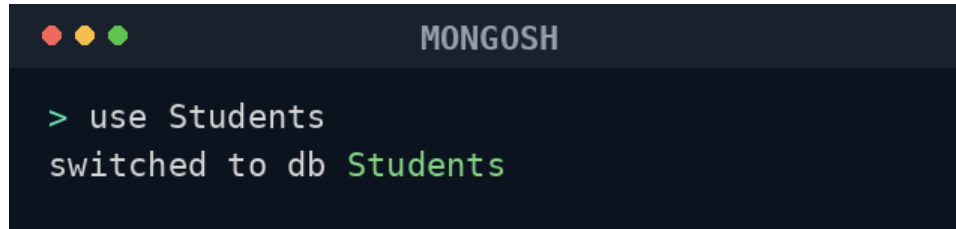
Section: BCADS23

1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Student database.

A screenshot of a terminal window titled 'MONGOSH'. The window has a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. The terminal shows the command '> use Students' and the response 'switched to db Students' where 'Students' is highlighted in green.

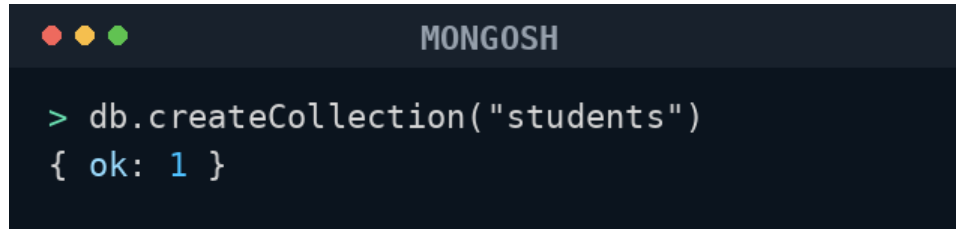
```
> use Students
switched to db Students
```

2. CREATE COLLECTION

MongoDB Query:

```
db.createCollection("students")
```

Explanation: Creates a collection named students.

A terminal window titled 'MONGOSH' with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal shows a command prompt '>' followed by the command 'db.createCollection("students")'. Below the command, the output is displayed as '{ ok: 1 }' in a light blue color.

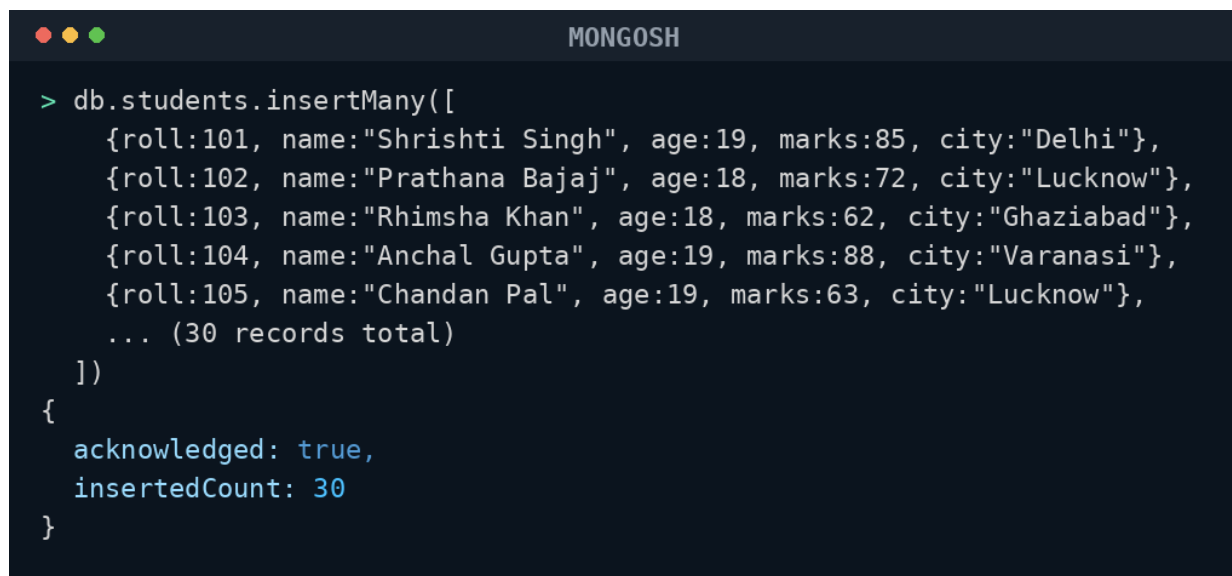
```
> db.createCollection("students")
{ ok: 1 }
```

3. INSERT 30 STUDENTS

MongoDB Query:

```
db.students.insertMany([
  {roll:101, name:"Shrishti Singh", age:19, marks:85, city:"Delhi"},
  {roll:102, name:"Prathana Bajaj", age:18, marks:72, city:"Lucknow"},
  {roll:103, name:"Rhimsha Khan", age:18, marks:62, city:"Ghaziabad"},
  {roll:104, name:"Anchal Gupta", age:19, marks:88, city:"Varanasi"},
  {roll:105, name:"Chandan Pal", age:19, marks:63, city:"Lucknow"},
  {roll:106, name:"Ishika Shrivastava", age:19, marks:83, city:"Prayagraj"},
  {roll:107, name:"Kritika Verma", age:18, marks:65, city:"Ghaziabad"},
  {roll:108, name:"Saksham Rao", age:19, marks:71, city:"Ghaziabad"},
  {roll:109, name:"Devyani Joshi", age:19, marks:93, city:"Ghaziabad"},
  {roll:110, name:"Aryan Malhotra", age:19, marks:61, city:"Varanasi"},
  {roll:111, name:"Nikhil Ali", age:20, marks:54, city:"Ghaziabad"},
  {roll:112, name:"Rohan Tripathi", age:20, marks:95, city:"Prayagraj"},
  {roll:113, name:"Tanisha Kapoor", age:19, marks:57, city:"Delhi"},
  {roll:114, name:"Faizan Ahmed", age:19, marks:85, city:"Prayagraj"},
  {roll:115, name:"Vanshika Rawat", age:20, marks:60, city:"Lucknow"},
  {roll:116, name:"Lakshay Sethi", age:18, marks:61, city:"Noida"},
  {roll:117, name:"Muskan Jaiswal", age:19, marks:84, city:"Varanasi"},
  {roll:118, name:"Harshvardhan Yadav", age:20, marks:90, city:"Ghaziabad"},
  {roll:119, name:"Gungun Mishra", age:20, marks:94, city:"Lucknow"},
  {roll:120, name:"Sarthak Bhatt", age:20, marks:61, city:"Prayagraj"},
  {roll:121, name:"Alisha Khan", age:20, marks:95, city:"Noida"},
  {roll:122, name:"Divyanshu Pandey", age:19, marks:93, city:"Lucknow"},
  {roll:123, name:"Kashish Arora", age:20, marks:94, city:"Delhi"},
  {roll:124, name:"Rajat Chauhan", age:19, marks:81, city:"Delhi"},
  {roll:125, name:"Shivangi Dubey", age:20, marks:61, city:"Ghaziabad"},
  {roll:126, name:"Yashwant Rana", age:18, marks:60, city:"Delhi"},
  {roll:127, name:"Bhavya Saxena", age:19, marks:73, city:"Ghaziabad"},
  {roll:128, name:"Nitya Agarwal", age:18, marks:63, city:"Delhi"},
  {roll:129, name:"Om Prakash", age:19, marks:95, city:"Ghaziabad"},
  {roll:130, name:"Tanmay Srivastava", age:19, marks:72, city:"Varanasi"}
])
```

Explanation: Inserts multiple student documents into the collection.



```
> db.students.insertMany([
  {roll:101, name:"Shrishti Singh", age:19, marks:85, city:"Delhi"},
  {roll:102, name:"Prathana Bajaj", age:18, marks:72, city:"Lucknow"},
  {roll:103, name:"Rhimsha Khan", age:18, marks:62, city:"Ghaziabad"},
  {roll:104, name:"Anchal Gupta", age:19, marks:88, city:"Varanasi"},
  {roll:105, name:"Chandan Pal", age:19, marks:63, city:"Lucknow"},
  ... (30 records total)
])
{
  acknowledged: true,
  insertedCount: 30
}
```

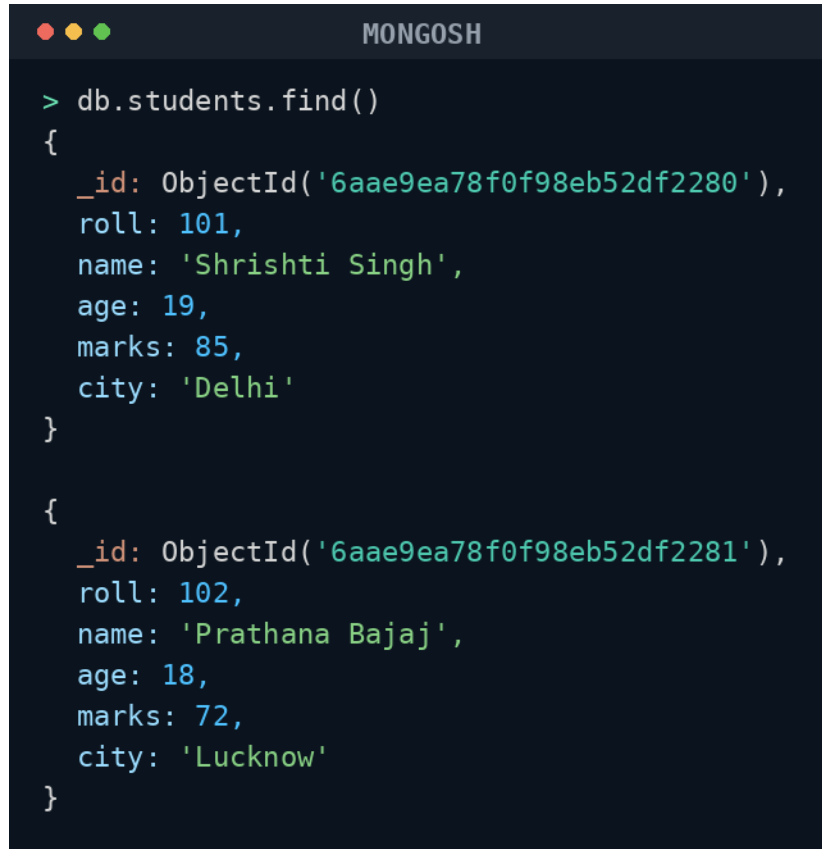
All 30 student records were inserted successfully into the 'students' collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

`db.students.find()`

Explanation: Retrieves documents from the students collection.



```
MONGOSH
> db.students.find()
{
  _id: ObjectId('6aae9ea78f0f98eb52df2280'),
  roll: 101,
  name: 'Shrishti Singh',
  age: 19,
  marks: 85,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2281'),
  roll: 102,
  name: 'Prathana Bajaj',
  age: 18,
  marks: 72,
  city: 'Lucknow'
}
```

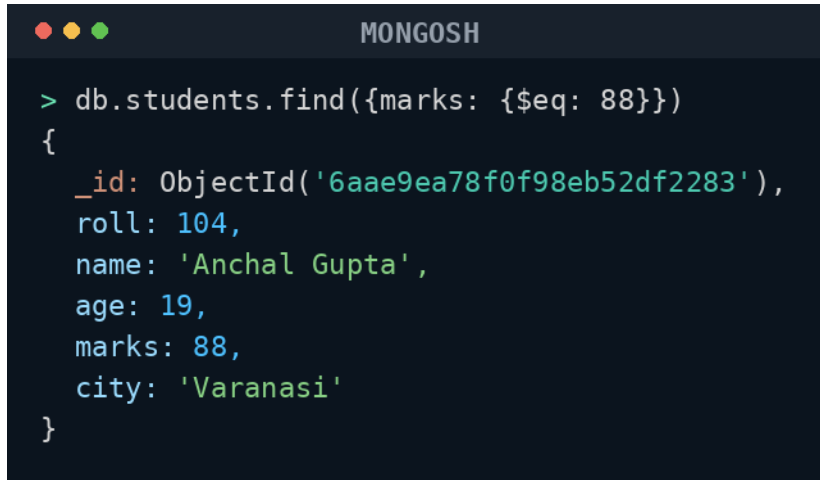
Showing the first 2 of 30 documents returned by this query.

5. EQUAL TO — \$eq

MongoDB Query:

```
db.students.find({marks: {$eq: 88}})
```

Explanation: Finds students whose marks are exactly 88.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({marks: {$eq: 88}})`. The output is a JSON document representing a student: `{ _id: ObjectId('6aae9ea78f0f98eb52df2283'), roll: 104, name: 'Anchal Gupta', age: 19, marks: 88, city: 'Varanasi' }`.

```
MONGOSH
> db.students.find({marks: {$eq: 88}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2283'),
  roll: 104,
  name: 'Anchal Gupta',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.students.find({marks: {$gt: 85}})
```

Explanation: Finds students whose marks are greater than 85.

```
MONGOSH

> db.students.find({marks: {$gt: 85}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2283'),
  roll: 104,
  name: 'Anchal Gupta',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2288'),
  roll: 109,
  name: 'Devyani Joshi',
  age: 19,
  marks: 93,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228b'),
  roll: 112,
  name: 'Rohan Tripathi',
  age: 20,
  marks: 95,
  city: 'Prayagraj'
}
```

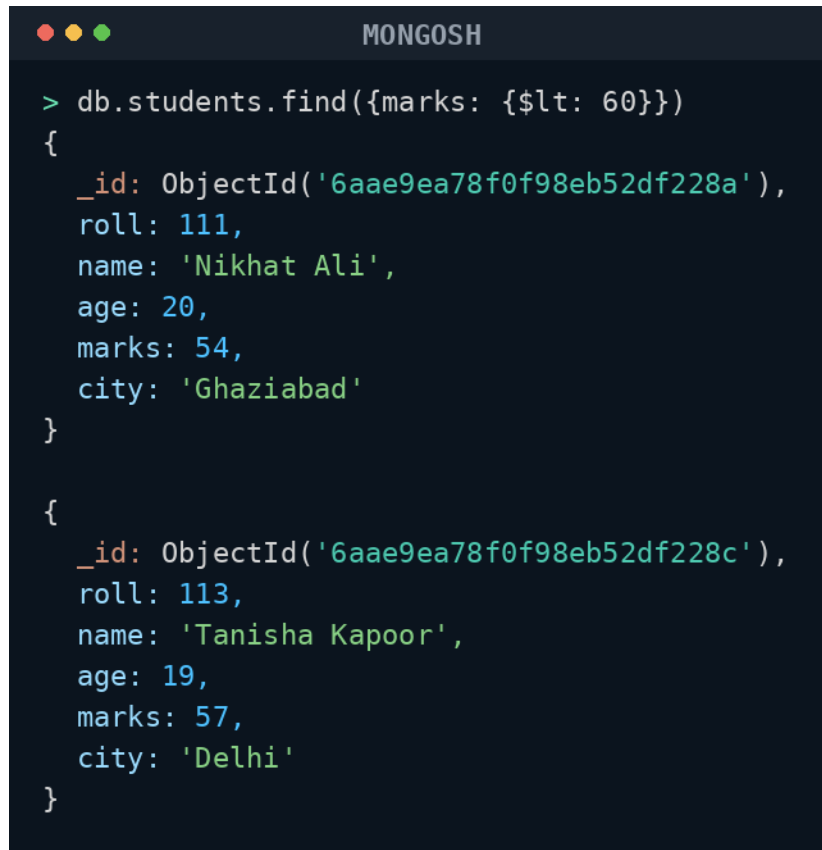
Showing 3 of 9 matching documents.

7. LESS THAN — \$lt

MongoDB Query:

`db.students.find({marks: {$lt: 60}})`

Explanation: Finds students whose marks are less than 60.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({marks: {$lt: 60}})`. The output consists of two JSON documents. The first document represents a student with `_id: ObjectId('6aae9ea78f0f98eb52df228a')`, `roll: 111`, `name: 'Nikhat Ali'`, `age: 20`, `marks: 54`, and `city: 'Ghaziabad'`. The second document represents a student with `_id: ObjectId('6aae9ea78f0f98eb52df228c')`, `roll: 113`, `name: 'Tanisha Kapoor'`, `age: 19`, `marks: 57`, and `city: 'Delhi'`.

```
> db.students.find({marks: {$lt: 60}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df228a'),
  roll: 111,
  name: 'Nikhat Ali',
  age: 20,
  marks: 54,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228c'),
  roll: 113,
  name: 'Tanisha Kapoor',
  age: 19,
  marks: 57,
  city: 'Delhi'
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.students.find({marks: {$gte: 90}})
```

Explanation: Finds students whose marks are 90 or above.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({marks: {$gte: 90}})`. The output displays three JSON documents. Each document has fields for `_id` (an ObjectId), `roll`, `name`, `age`, `marks`, and `city`. The first document is for Devyani Joshi (roll 109, age 19, marks 93, city Ghaziabad). The second is for Rohan Tripathi (roll 112, age 20, marks 95, city Prayagraj). The third is for Harshvardhan Yadav (roll 118, age 20, marks 90, city Ghaziabad).

```
> db.students.find({marks: {$gte: 90}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2288'),
  roll: 109,
  name: 'Devyani Joshi',
  age: 19,
  marks: 93,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228b'),
  roll: 112,
  name: 'Rohan Tripathi',
  age: 20,
  marks: 95,
  city: 'Prayagraj'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2291'),
  roll: 118,
  name: 'Harshvardhan Yadav',
  age: 20,
  marks: 90,
  city: 'Ghaziabad'
}
```

Showing 3 of 8 matching documents.

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.students.find({marks: {$lte: 70}})
```

Explanation: Finds students whose marks are 70 or below.

```
MONGOSH

> db.students.find({marks: {$lte: 70}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2282'),
  roll: 103,
  name: 'Rhimsha Khan',
  age: 18,
  marks: 62,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),
  roll: 105,
  name: 'Chandan Pal',
  age: 19,
  marks: 63,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2286'),
  roll: 107,
  name: 'Kritika Verma',
  age: 18,
  marks: 65,
  city: 'Ghaziabad'
}
```

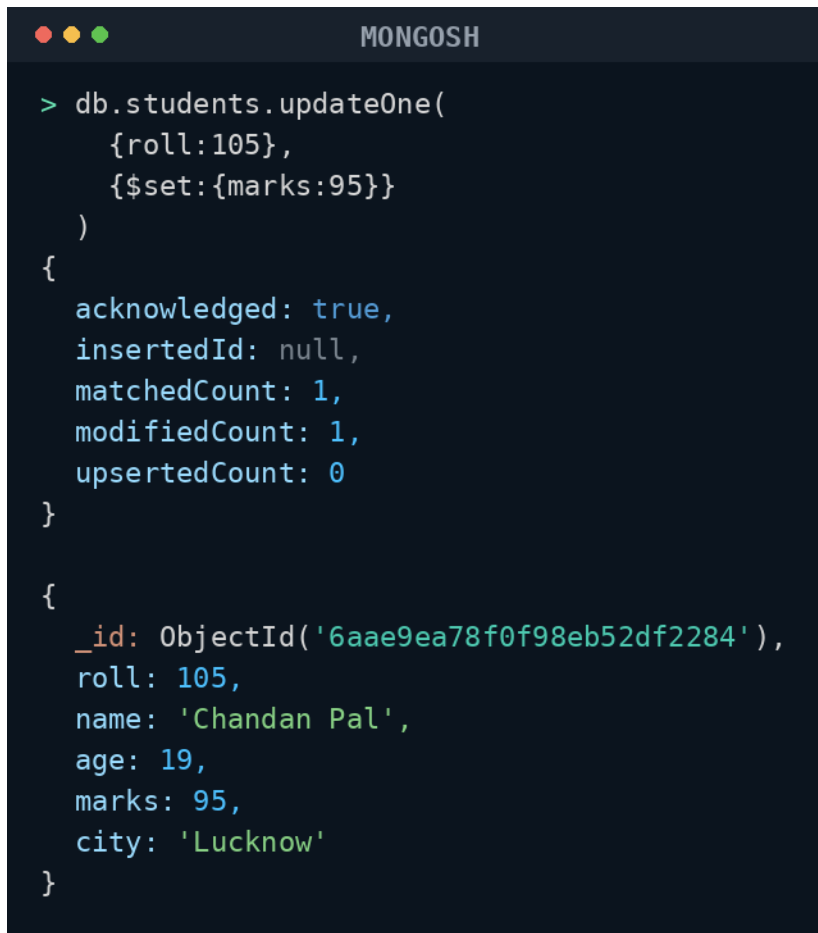
Showing 3 of 12 matching documents.

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(  
  {roll:105},  
  {$set:{marks:95}}  
)
```

Explanation: Updates the marks to 95 of the student with roll number 105.



```
MONGOSH  
  
> db.students.updateOne(  
  {roll:105},  
  {$set:{marks:95}}  
)  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
  
{  
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),  
  roll: 105,  
  name: 'Chandan Pal',  
  age: 19,  
  marks: 95,  
  city: 'Lucknow'  
}
```

11. DELETE ONE OPERATION

MongoDB Query:

db.students.deleteOne({roll:110})

Explanation: Deletes one document matching roll number 110.

A terminal window titled 'MONGOSH' with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal displays a MongoDB command and its output. The command is entered on the first line, and the output is shown on the following three lines.

```
> db.students.deleteOne({roll:110})
{
  acknowledged: true,
  deletedCount: 1
}
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({  
  marks: {$lt:60}  
})
```

Explanation: Deletes all students having marks below 60.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed:

```
> db.students.deleteMany({  
  marks: {$lt:60}  
})  
{  
  acknowledged: true,  
  deletedCount: 2  
}
```

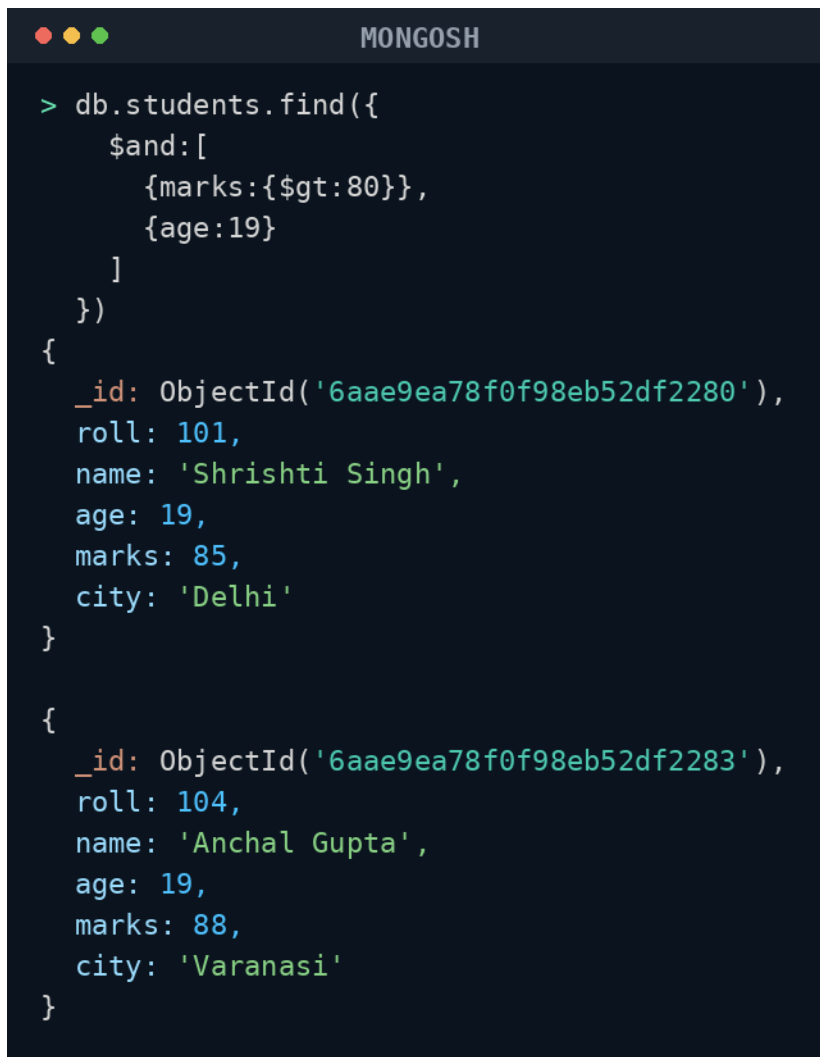
Removed roll numbers 111, 113 (marks below 60).

13. AND OPERATION — \$and

MongoDB Query:

```
db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({ $and:[ {marks:{$gt:80}}, {age:19} ] })`. The output displays two JSON documents. The first document has fields: `_id` (ObjectId), `roll` (101), `name` ('Shrishti Singh'), `age` (19), `marks` (85), and `city` ('Delhi'). The second document has fields: `_id` (ObjectId), `roll` (104), `name` ('Anchal Gupta'), `age` (19), `marks` (88), and `city` ('Varanasi').

```
MONGOSH
> db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2280'),
  roll: 101,
  name: 'Shrishti Singh',
  age: 19,
  marks: 85,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2283'),
  roll: 104,
  name: 'Anchal Gupta',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}
```

Showing 2 of 10 matching documents.

14. OR OPERATION — \$or

MongoDB Query:

```
db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Delhi"}
  ]
})
```

Explanation: Finds students from Lucknow or Delhi.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({ $or:[ {city:"Lucknow"}, {city:"Delhi"} ] })`. The output displays two JSON documents. The first document is for a student with `_id: ObjectId('6aae9ea78f0f98eb52df2280')`, `roll: 101`, `name: 'Shrishti Singh'`, `age: 19`, `marks: 85`, and `city: 'Delhi'`. The second document is for a student with `_id: ObjectId('6aae9ea78f0f98eb52df2281')`, `roll: 102`, `name: 'Prathana Bajaj'`, `age: 18`, `marks: 72`, and `city: 'Lucknow'`.

```
> db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Delhi"}
  ]
})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2280'),
  roll: 101,
  name: 'Shrishti Singh',
  age: 19,
  marks: 85,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2281'),
  roll: 102,
  name: 'Prathana Bajaj',
  age: 18,
  marks: 72,
  city: 'Lucknow'
}
```

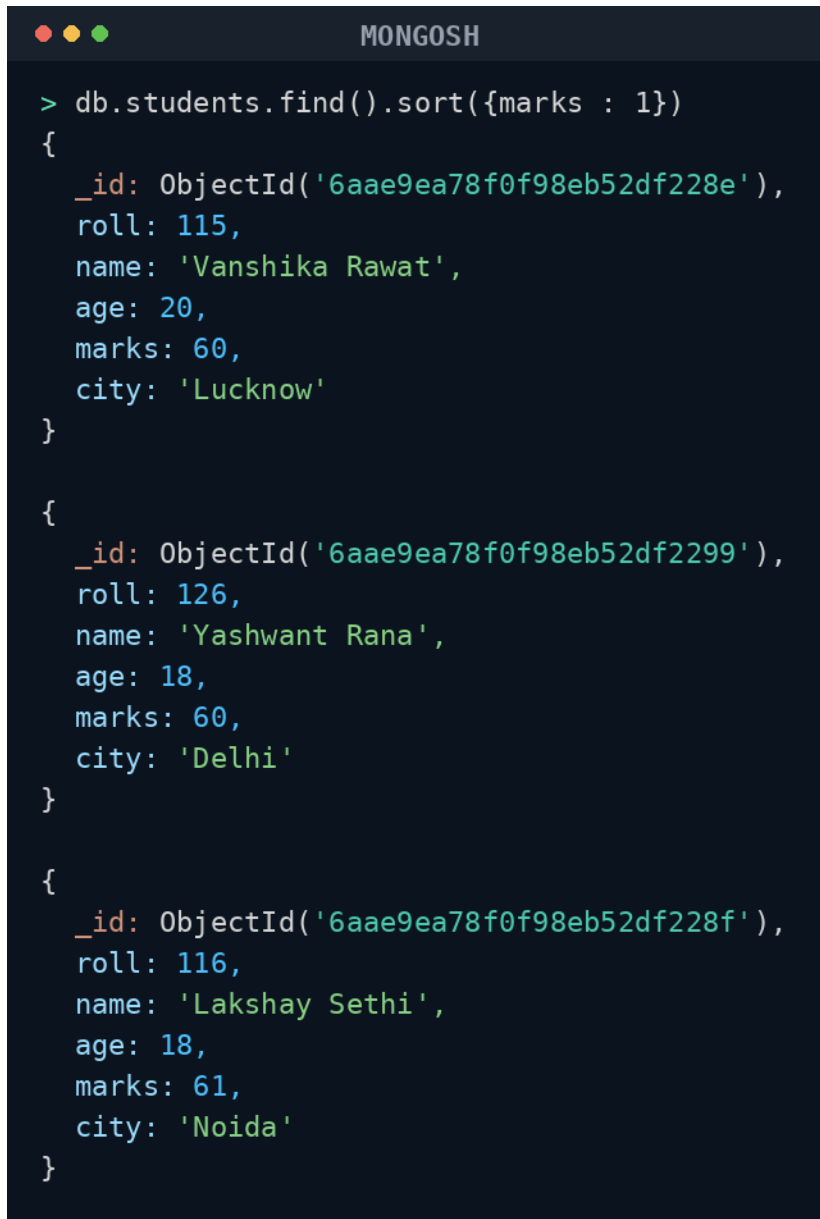
Showing 2 of 10 matching documents.

15. SORT OPERATION

A. MongoDB Query:

```
db.students.find().sort({marks : 1})
```

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

A screenshot of a MongoDB shell window titled 'MONGOSH'. The prompt is '> db.students.find().sort({marks : 1})'. The output shows three JSON documents sorted by the 'marks' field in ascending order. The first document has marks 60, the second has marks 60, and the third has marks 61. Each document contains fields for _id, roll, name, age, marks, and city.

```
> db.students.find().sort({marks : 1})
{
  _id: ObjectId('6aae9ea78f0f98eb52df228e'),
  roll: 115,
  name: 'Vanshika Rawat',
  age: 20,
  marks: 60,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2299'),
  roll: 126,
  name: 'Yashwant Rana',
  age: 18,
  marks: 60,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228f'),
  roll: 116,
  name: 'Lakshay Sethi',
  age: 18,
  marks: 61,
  city: 'Noida'
}
```

B. MongoDB Query:

```
db.students.find().sort({marks: -1})
```

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
MONGOSH

> db.students.find().sort({marks: -1})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),
  roll: 105,
  name: 'Chandan Pal',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228b'),
  roll: 112,
  name: 'Rohan Tripathi',
  age: 20,
  marks: 95,
  city: 'Prayagraj'
}

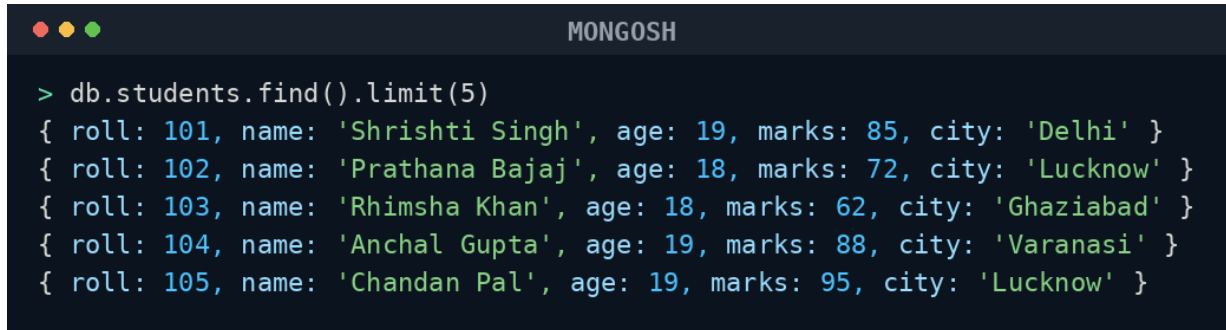
{
  _id: ObjectId('6aae9ea78f0f98eb52df2294'),
  roll: 121,
  name: 'Alisha Khan',
  age: 20,
  marks: 95,
  city: 'Noida'
}
```

16. LIMIT OPERATION

MongoDB Query:

`db.students.find().limit(5)`

Explanation: Displays only the first five documents.



```
> db.students.find().limit(5)
{ roll: 101, name: 'Shrishti Singh', age: 19, marks: 85, city: 'Delhi' }
{ roll: 102, name: 'Prathana Bajaj', age: 18, marks: 72, city: 'Lucknow' }
{ roll: 103, name: 'Rhimsha Khan', age: 18, marks: 62, city: 'Ghaziabad' }
{ roll: 104, name: 'Anchal Gupta', age: 19, marks: 88, city: 'Varanasi' }
{ roll: 105, name: 'Chandan Pal', age: 19, marks: 95, city: 'Lucknow' }
```

17. SEARCH BY CITY

MongoDB Query:

```
db.students.find({city:"Lucknow"})
```

Explanation: Finds all students whose city is Lucknow.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query and its results. The query is `> db.students.find({city:"Lucknow"})`. The output consists of three JSON documents, each representing a student from Lucknow. The first document has an `_id` of `ObjectId('6aae9ea78f0f98eb52df2281')`, a `roll` of `102`, a `name` of `'Prathana Bajaj'`, an `age` of `18`, `marks` of `72`, and a `city` of `'Lucknow'`. The second document has an `_id` of `ObjectId('6aae9ea78f0f98eb52df2284')`, a `roll` of `105`, a `name` of `'Chandan Pal'`, an `age` of `19`, `marks` of `95`, and a `city` of `'Lucknow'`. The third document has an `_id` of `ObjectId('6aae9ea78f0f98eb52df228e')`, a `roll` of `115`, a `name` of `'Vanshika Rawat'`, an `age` of `20`, `marks` of `60`, and a `city` of `'Lucknow'`.

```
> db.students.find({city:"Lucknow"})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2281'),
  roll: 102,
  name: 'Prathana Bajaj',
  age: 18,
  marks: 72,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),
  roll: 105,
  name: 'Chandan Pal',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228e'),
  roll: 115,
  name: 'Vanshika Rawat',
  age: 20,
  marks: 60,
  city: 'Lucknow'
}
```

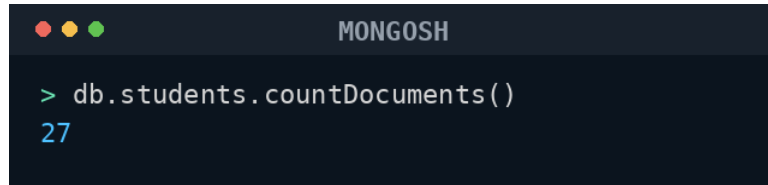
Showing 3 of 5 matching documents.

18. COUNT DOCUMENTS

MongoDB Query:

db.students.countDocuments()

Explanation: Counts documents currently present in the collection (after the delete operations).



```
MONGOSH
> db.students.countDocuments()
27
```

MONGODB OPERATORS USED:

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq:88}
\$gt	Greater than	marks: {\$gt:85}
\$lt	Less than	marks: {\$lt:60}
\$gte	Greater than or equal to	marks: {\$gte:90}
\$lte	Less than or equal to	marks: {\$lte:70}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]